Badrul Alam Siddique **Masfy**

Manager, Technology Division · BRAC International · Data Platforms & Delivery at Scale

Dhaka, Bangladesh

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Summary

Technology manager with 5.5+ years across system architecture, database engineering, ETL/data pipelines and offline-first platforms. Experienced in leading delivery for multi-country programmes: requirements, solution design, access governance, UAT and rollout. Led engineering for **BInsight**, **BRAC International**'s MEL platform serving 30,000+ field users, and cut report generation on 70M+ row PostgreSQL tables from 5–10 minutes to under 1 minute. Currently Manager, Technology Division at BRAC International.

Core Competencies

Technology & Delivery Leadership
- Roadmap ownership, milestones, risk/issue logs
- Requirements, stakeholder communication, solution design
- Leading cross-functional teams; mentoring engineers
- Release coordination, UAT, phased rollout
- Change management, training, user adoption
- Agile/Scrum delivery in multi-country INGO contexts

Architecture & Access Governance
- Scalable backend and configuration-driven platform design
- RBAC and segregation-of-duties design
- Approval workflows; minimum-necessary access
- REST API design, service architecture, system integrations

Database Architecture & Performance
- PostgreSQL partitioning at 70M+ row scale
- Query/index optimisation; data modelling
- PostgreSQL ↔ SQLite offline sync
- PostgreSQL, MS SQL Server, SQLite, MongoDB, Firebase

Data Pipelines, Reporting & BI
- ETL pipelines: Airflow, Airbyte, dbt
- Data transformation, validation and reconciliation
- Power BI dashboards, KPI and executive reporting

Monitoring, Evaluation & Learning (M&E)
- MEL platform leadership, 30,000+ field users
- Offline-first data collection; survey and workflow design
- Data-quality, validation and decision-ready reporting

Engineering Background, DevOps & Tools
- C#, .NET Core, Node.js, Java; Flutter/Dart; ReactJS (familiar)
- AI-assisted engineering (OpenAI API, Claude MCP)
- Docker, AWS, Jenkins CI/CD, Grafana, Git, Jira

Work Experience

BRAC International
Dhaka, Bangladesh

Manager, Technology Division — *Nov. 2026 – Present*

- Lead technology strategy and roadmap across BRAC International's multi-country programmes, aligning IT initiatives with organisational digitisation priorities.
- Manage end-to-end delivery of technology initiatives, from requirement discovery and solution design to UAT, release and stabilisation, working with internal teams, partners and vendors.
- Align data platforms, access governance and KPI reporting with organisational and country-office priorities.
- Own system integration and data-exchange design between source and target systems, balancing quality, timelines and value for money.
- Drive cross-functional collaboration with head-office and country-office stakeholders, applying change management, training and adoption practices.

Streams Tech Ltd
Dhaka, Bangladesh (Remote)

Software Engineer · Technical Lead, BInsight Platform — *Mar. 2024 – Nov. 2026*

- Served as the primary technical point of contact between Streams Tech and BRAC International for BInsight, running recurring planning, requirement-gathering and review meetings with HO and country-office stakeholders across multi-country operations.
- Led a cross-functional team across BInsight and an internal ERP; owned the BInsight roadmap, milestones and risk/issue log; translated business-architecture decisions into system design; coordinated releases, UAT and post-go-live stabilisation.
- Led PostgreSQL LIST partitioning and schema normalisation across 70M+ row tables, cutting KPI-dashboard and report generation from 5–10 min to under 1 min; owned ETL strategy (Airflow, Airbyte, dbt) for 3x faster warehouse queries.
- Designed RBAC and approval-workflow structures across BInsight and an internal ERP, enforcing segregation of duties and minimum-necessary-access governance across country, HQ and departmental roles.
- Led the end-to-end build of an internal ERP suite (HR, in-house issue tracker, leave management), turning BInsight's core engine into a company-wide platform delivered by a small team within months.
- Directed adoption of an AI-assisted workflow configurator (OpenAI API, Claude MCP), cutting configuration setup time by 60%; directed a .NET Core and Grafana telemetry pipeline covering 20+ production deployments.
- **Tools:** C#, .NET Core, PostgreSQL, MS SQL Server, SQLite, Flutter, Airflow, Airbyte, dbt, Power BI, Grafana, Git, Jira.

Streams Tech Ltd

Dhaka, Bangladesh

Associate Software Engineer · Platform Engineering — *Mar. 2023 – Mar. 2024*

- Directed development of an offline-first Flutter application with bidirectional PostgreSQL ↔ SQLite sync (Riverpod), supporting 30,000+ field users with zero data loss on reconnection.
- Architected and led delivery of a backend-driven workflow/configuration engine letting non-developers self-configure 140+ settings without redeployment, cutting deployment downtime by 70%.
- Coordinated readiness assessments and phased rollout planning with country teams ahead of each release.
- Led end-to-end delivery of a full-stack property management platform (.NET Core APIs + Flutter/BLoC), from architecture through rollout.
- **Tools:** C#, .NET Core, Flutter, Dart, PostgreSQL, MS SQL Server, Git, Jira.

Innovex IT

Sylhet, Bangladesh

Junior Software Engineer — *Mar. 2022 – Mar. 2023*

- Built an inventory management system (Flutter + Node.js + MongoDB) covering product, sales, customer, finance and client management.
- Reduced app network usage by 40% on low-end devices through API response compression, lazy loading and adaptive rendering.
- Set up Jenkins CI/CD pipelines, cutting deployment time from 30 min to 5 min.
- **Tools:** Flutter, Dart, Node.js, Firebase, MongoDB, Git.

Notable Projects

BInsight

BRAC International

Enterprise MEL Platform | Tools: PostgreSQL, .NET Core, Flutter, Airflow, dbt, Power BI — *Mar. 2023 – Nov. 2026*

- Database and delivery lead for BRAC International's flagship MEL platform serving 30,000+ field users; owned the database layer (70M+ rows), ETL pipelines, REST APIs and offline sync engine.
- Modules: registration, event management, survey engine, GIS integration, cash disbursement, real-time analytics and reporting.

HR, Issue Tracking & Leave Management

Streams Tech Ltd

Internal ERP Suite | Tools: .NET Core, PostgreSQL — *2024 – Nov. 2026*

- Captured requirements, designed the data model and RBAC/segregation-of-duties structure, and shipped an in-house issue tracker (Jira alternative) and leave management with approval workflows, used company-wide.

Natural-language configuration

Streams Tech Ltd

AI-Assisted Workflow Configurator | Tools: OpenAI API, Claude MCP, .NET Core — *2024 – Nov. 2026*

- Generates mobile UI configs and survey schemas from natural-language input; cut configuration setup time by 60%.

Flat Suite

Streams Tech Ltd

Mobile Application | Tools: Flutter (BLoC), .NET Core — *Oct. 2023 – Feb. 2024*

- Led delivery of a property management app with expense tracking, charge collection, role-based access for residents and board members, financial reporting and automated reconciliation.

DokanE

Innovex IT

Full Stack Application | Tools: Node.js, Flutter, Firebase, MongoDB — *Oct. 2022 – Feb. 2023*

- Built Node.js + MongoDB backend for inventory management; Flutter frontend with BLoC pattern, lazy loading, and memory-efficient data serialization for low-end device support.

E-commerce, Seller, and Bank Integration

Undergrad

Web Application | Tools: Node.js, ReactJS, MongoDB — *Feb. 2022 – Jun. 2022*

- Implemented three-way REST API integration across E-commerce, Supplier, and Bank services for real-time transaction confirmation, inventory supply, and balance management.

PocketLencer

Undergrad

Cross Platform Application | Tools: Node.js, Flutter, Dart, Firebase — *Aug. 2021 – Nov. 2021*

- Built iOS and Android app from a single Flutter codebase using MVP architecture and BLoC state management for an original artwork marketplace.

Education

Shahjalal University of Science and Technology

Sylhet, Bangladesh

B.Sc(Eng). in Computer Science and Engineering — *Jan. 2018 – Feb. 2023*

- GPA: 3.55 / 4.00
- Notable Coursework: Data Structure, Algorithms, AI, Design Patterns, RDBMS, Computer Architecture